Tutorial 6 (Friday)

4:21 PM

Problem #2: Check which one of the following functions is a solution to the differential equation $y'' + y = \cos x$.

(A) $\frac{1}{2}(e^x - \cos x)$ (B) $\frac{1}{2}(e^x + \cos x)$ (C) $\frac{1}{2}(\cos x + x \sin x)$ (D) $\frac{1}{2}(\sin x + x \cos x)$ (E) $\frac{1}{2}(\sin x - x \cos x)$ (F) $\frac{1}{2}(e^x + \sin x)$

Goal: guess and check!

Try (A): $y = \frac{1}{2}e^x - \frac{1}{2}\cos(x)$

$y' = \frac{1}{2}e^x + \frac{1}{2}\sin(x)$

$y'' = \frac{1}{2}e^x + \frac{1}{2}\cos(x)$

$\Rightarrow y + y'' = e^x \neq \cos(x)$

Notice: e^x is hard to eliminate. So, likely B & F also fail!

Try (C): $y = \frac{1}{2}\cos(x) + \frac{1}{2}x\sin(x)$

$y' = -\frac{1}{2}\sin(x) + \frac{\sin(x)}{2} + \frac{x}{2}\cos(x)$

$y'' = -\frac{1}{2}\cos(x) + \frac{\cos(x)}{2} + \frac{\cos(x)}{2} - \frac{x}{2}\sin(x)$

$y + y'' = \cos(x)$ ✓

Problem #3: For what two values of r does the function $y = e^{rx}$ satisfy the differential equation $y'' + 2y' - 35y = 0$?

Note: $y' = re^{rx}$ $y'' = r^2e^{rx}$

So, $r^2e^{rx} + 2re^{rx} - 35e^{rx} = 0$

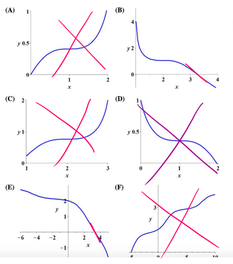
$e^{rx}(r^2 + 2r - 35) = 0$ ($e^{rx} > 0$)

$e^{rx}(r + 7)(r - 5) = 0$

$r = -7, 5$

Problem #4: By inspecting the following differential equation and its properties, determine which of the given graphs could be the graph of a solution to the equation.

$y' = -e^{-y} + \sin x$



Note: $e^{-x} > 0$

$\Rightarrow -e^{-x} < 0 \rightarrow y' < 0 \rightarrow$ decreasing!

When $x = \pi \approx 3.14$ and $y = 0$,

$y' = -e^{0 + \sin(\pi)} = -e^0 = -1$

slope of -1. looks like (E)!

Problem #5: Let $y(x)$ be the solution to the following initial value problem.

$\frac{dy}{dx} = \frac{(\ln x)^4}{xy^2}$, $y(1) = 5$

Find $y(e)$.

$\int y^2\,dy = \int \frac{(\ln x)^4}{x}\,dx$

$u = \ln(x)$

$du = \frac{1}{x}dx$ $\int u^4\,du$

$= \frac{u^5}{5} + C$

$\frac{y^3}{3} = \frac{(\ln(x))^5}{5} + C$

$y = \left(\frac{3(\ln x)^5}{5} + C_1\right)^{1/3}$

$y(1) = (C_1)^{1/3} = 5 \Rightarrow C_1 = 125$

Hence, $y(e) = \left(\frac{3}{5} + 125\right)^{1/3}$ ← ok for childsmath

Problem #6: Find the orthogonal trajectories of the family of curves $y^4 = kx^5$.

(A) $3y^3 + \frac{5}{2}x^3 = C$ (B) $\frac{5}{2}y^3 + 2x^2 = C$ (C) $\frac{5}{2}y^3 + \frac{5}{2}x^3 = C$ (D) $2y^2 + 2x^2 = C$ (E) $3y^3 + 2x^2 = C$ (F) $\frac{7}{2}y^2 + \frac{7}{2}x^3 = C$ (G) $\frac{7}{2}y^3 + 3x^2 = C$ (H) $\frac{5}{2}y^2 + \frac{3}{2}x^2 = C$

$4y^3\frac{dy}{dx} = 5kx^4$ $\left(u = \frac{y^4}{x^5}\right)$

$\frac{dy}{dx} = \frac{5kx^4}{4y^3} = \frac{5\left(\frac{y^4}{x^5}\right)x^4}{4y^3} = \frac{5y}{4x} \Rightarrow \frac{-4x}{5y}$

$\Rightarrow \int 5y\,dy = \int -4x\,dx$

$\frac{5y^2}{2} = -\frac{4x^2}{2} + c \Rightarrow \frac{5y^2}{2} + 2x^2 = C$

Problem #7: A sample of tritium-3 decayed to 94% of its original amount after 2 years. How long would it take the sample (in years) to decay to 20% of its original amount?

use: $y(t) = y(0)e^{kt}$

Know: $y(2) = 94\%$ $y(0) = 100\% = 1$

$0.94 = e^{k(2)} \Rightarrow \frac{\ln(0.94)}{2} = k$

WTFind: $0.20 = e^{kt}$

$\Rightarrow \frac{\ln(0.2)}{k} = t$

$\Rightarrow \frac{\ln(0.1)2}{\ln(0.94)} = t$

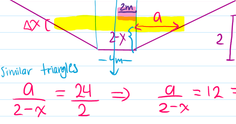
convert "symbolically":

$t = \frac{2\ln(3/10)}{\ln(94/100)}$

Problem #1: A gate in an irrigation canal is constructed in the form of a trapezoid 4 m wide at the bottom, 52 m wide at the top, and 2 m high. It is placed vertically in the canal so that the water just covers the gate. Find the hydrostatic force on one side of the gate.

Note that your answer should be in Newtons, and use $g = 9.8\ m/s^2$.

Not to scale:



$x_i^* \in [x_{i-1}, x_i]$

Similar triangles

$\frac{a}{2-x} = \frac{24}{2} \Rightarrow \frac{a}{2-x} = 12 \Rightarrow a = 24 - 12x$

$w_i = 2(2 + a) = 2(2 + 24 - 12x) = 2(26 - 12x) = 52 - 24x$

$A_i \approx w\Delta x = (52 - 24x)\Delta x$

$P_i \approx 1000gx_i$

$F_i = P_iA_i \approx 1000gx_i(52 - 24x_i)\Delta x$

$F = \lim_{n\to\infty} \sum_{i=1}^{n} 1000gx_i(52 - 24x_i)\Delta x$

$= \int_0^2 1000gx(52 - 24x)\,dx$

$= 1000g\int_0^2 52x - 24x^2\,dx$

$= 1000g\left(\frac{52x^2}{2} - \frac{24x^3}{3}\right)\Big|_0^2$

$= 1000g\left(26(2^2) - 8(2^3)\right)$

$= 1000g(104 - 64)$

$= 40000(9.8) = 392000\ N$

(Tbh I never did physics... I'm a practicing statistician >:( )